

TSUYAMA, Japan

WMO: 477560

Lat: 35.0648N

Lon: 134.0090E

Elev: 148

StdP: 99.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.9	-7.2	2.1	1.3	-6.1	2.3	1.7	8.3	2.9	7.2	3.3	1.6	270	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	33.6	25.2	32.4	24.9	31.1	24.5	26.2	31.3	25.6	30.4	25.1	29.5	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.4	28.1	24.4	19.7	27.8	23.9	19.1	27.5	82.5	31.6	80.0	30.5	77.8	29.7	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.5	5.3	4.4	DB	-6.1	35.3	1.6	1.1	-7.3	36.0	-8.2	36.7	-9.1	37.3	-10.3	38.1	
			WB	-6.5	27.2	1.6	0.5	-7.7	27.6	-8.6	27.9	-9.5	28.2	-10.7	28.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.9	3.9	7.2	12.7	17.9	21.8	26.0	27.1	22.8	16.6	10.3	5.0
	DBStd	8.79	2.12	2.66	3.06	3.37	2.47	2.36	2.26	1.89	2.90	3.05	3.21	2.58
	HDD10.0	696	221	173	97	13	0	0	0	0	0	1	36	157
	HDD18.3	2167	479	405	344	170	38	2	0	0	2	71	242	414
	CDD10.0	2364	0	1	11	94	245	355	494	529	384	206	44	1
	CDD18.3	792	0	0	0	1	24	107	236	271	136	17	0	0
	CDH23.3	6348	0	0	0	37	245	638	1968	2521	868	71	0	0
	CDH26.7	2178	0	0	0	4	38	144	730	1021	237	5	0	0
Wind	WSAvg	1.7	1.9	1.9	2.0	1.9	1.8	1.6	1.6	1.6	1.6	1.5	1.5	1.8
Precipitation	PrecAvg	1500	90	83	107	109	147	171	208	121	177	107	91	91
	PrecMax	2017	148	119	193	181	309	299	378	336	437	239	202	148
	PrecMin	1037	50	47	57	34	56	47	41	40	55	38	31	32
	PrecStd	213	31	24	34	45	67	59	92	75	99	57	43	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	16.2	20.9	26.9	29.6	31.6	35.0	35.4	32.5	27.1	21.0	14.6
		MCWB	7.3	10.1	12.4	15.9	18.2	23.0	25.9	25.6	24.3	21.1	15.5	10.6
	2%	DB	9.9	13.1	18.1	24.0	27.6	29.7	33.3	34.0	30.7	25.1	19.1	12.9
		MCWB	6.5	8.3	11.0	14.5	17.5	21.7	25.5	25.3	23.6	19.1	14.5	9.4
	5%	DB	8.5	10.9	15.8	21.9	26.1	28.3	32.0	32.9	29.3	23.7	17.4	11.3
		MCWB	5.3	7.0	9.8	13.8	17.1	21.0	25.1	25.0	22.9	18.2	13.6	8.0
	10%	DB	7.2	9.2	13.6	19.9	24.4	27.0	30.7	31.7	27.9	22.2	15.9	9.8
		MCWB	4.3	5.8	9.2	12.9	16.5	20.7	24.7	24.8	22.1	17.3	12.7	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	12.4	14.2	18.6	21.5	24.9	27.0	27.1	25.7	22.9	17.6	12.5
		MCDB	10.3	14.9	17.4	21.8	24.4	28.7	32.5	32.6	30.3	25.2	19.0	13.2
	2%	WB	7.1	9.4	12.7	16.9	20.0	23.9	26.2	26.3	24.8	21.1	15.8	10.0
		MCDB	8.9	11.6	15.4	20.4	23.2	27.2	31.4	31.6	28.7	23.6	17.6	11.9
	5%	WB	5.7	7.6	11.3	15.6	19.0	23.0	25.6	25.8	24.1	19.6	14.5	8.5
		MCDB	7.9	10.2	14.4	19.4	22.6	25.9	30.4	30.7	27.4	22.0	16.6	10.6
	10%	WB	4.6	5.9	9.5	14.4	18.0	22.1	25.1	25.3	23.4	18.2	13.1	7.4
		MCDB	6.9	8.8	13.3	18.0	22.3	25.0	29.5	29.9	26.3	20.9	15.3	9.5
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.4	11.1	12.4	11.7	9.1	8.0	9.0	8.8	9.8	9.6	8.5
		MCDBR	10.4	12.8	15.3	16.2	15.4	11.7	10.1	10.8	10.1	11.2	10.9	10.5
		MCWBR	6.9	8.0	8.7	7.9	6.8	4.9	3.9	3.8	4.1	5.6	6.5	6.9
	5% WB	MCDBR	9.5	10.9	12.1	11.4	10.5	8.0	9.0	9.2	8.0	8.0	8.7	9.3
		MCWBR	6.8	7.7	8.2	6.5	5.9	4.0	3.8	3.6	3.6	4.8	5.9	6.8
Clear-Sky Solar Irradiance	taub	0.421	0.461	0.529	0.554	0.555	0.576	0.576	0.547	0.496	0.451	0.429	0.413	
	taud	2.037	1.957	1.821	1.819	1.873	1.893	1.952	2.024	2.123	2.151	2.113	2.081	
	Ebn at Noon	742	759	742	749	756	737	734	748	766	764	732	722	
	Edh at Noon	132	160	200	210	201	197	185	169	146	131	122	120	
All-Sky Solar Radiation	RadAvg	2.00	2.67	3.73	4.76	5.23	4.79	4.90	5.08	3.93	3.22	2.30	1.81	
	RadStd	0.15	0.25	0.31	0.41	0.49	0.40	0.69	0.55	0.37	0.30	0.19	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page